

Stock and Arrows Investing App



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4 AI Applications

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1 Introduction

1.1 Motivation

The stock market has traditionally been understood and analyzed primarily through economic indicators, company fundamentals, trading volumes, historical price patterns, and macroeconomic data. However, in reality, the forces that move stock prices extend far beyond economics alone. Political developments, statements or actions by prominent businesspeople, decisions made by politicians, wars, and other sudden, unpredictable events all play a significant role in shaping how a stock behaves. This reality is what shaped the motivation behind this project. As the team observed the current landscape of financial markets, it became clear that relying solely on conventional, purely economic or historical data was insufficient to capture the full picture of what drives stock price movement. This raised the central question that motivated the project: how can we arrive at a better, more complete approach to stock analysis and prediction, one that accounts not just for the numbers, but for the broader context surrounding them?

Given this reality, the team began exploring how to improve the way stock prediction and analysis is approached. The conclusion reached was that the most effective path forward was to incorporate news data — specifically, news that has a clear and measurable impact on stock prices — into the analytical process. By collecting this news and subjecting it to sentiment analysis, the aim was to better understand and anticipate how a given stock is likely to move over the coming days, and to what extent recent news coverage suggests whether the stock's price is likely to rise or fall. In essence, the problem being addressed is that traditional prediction systems, which rely on stock data alone, are incomplete on their own; they do not sufficiently account for the sentiment and implications carried by real-world news events, which can meaningfully shift market behavior in ways that pure historical or numerical analysis cannot capture.

1.2 Problem Statement

- Traditional stock prediction systems rely primarily on historical price data, trading volumes, and economic/fundamental indicators, treating the market as if it moves according to numbers alone.
- In reality, stock prices are heavily influenced by non-economic factors such as political events, decisions and statements by public figures and politicians, wars, and other sudden, unpredictable occurrences which these traditional systems largely fail to account for.
- News events that carry clear market moving implications are not systematically captured or incorporated into most existing prediction models, leaving a significant gap between what actually drives price movement and what the models are able to detect.
- Without a way to measure the sentiment and implied impact of real world news, predictions based purely on historical/numerical data risk being incomplete or misleading, especially during periods of high political, economic, or geopolitical volatility.
- This creates a clear need for a system that can bridge this gap one that combines conventional stock data analysis with sentiment extracted from relevant news, in order to produce predictions that are more reflective of real market conditions and more trustworthy for decision-making purposes.

1.3 Goals

1. Build a system that performs comprehensive analysis and prediction for a large-scale universe of stocks covering more than 15,000 stocks across more than 5 stock exchanges
2. Preserve and build upon the value of traditional, well-established stock analysis and prediction methods, rather than replacing them
3. Collect and process news data that has a demonstrable, measurable impact on stock prices, drawing from relevant financial and market-related sources
4. Apply sentiment analysis to this news data in order to gauge its likely direction of impact whether it suggests a stock's price is likely to rise or fall in the near term
5. Combine traditional stock-based prediction with news-driven sentiment analysis into a single, unified prediction mechanism, rather than treating them as separate, disconnected outputs
6. Use the sentiment-based layer to reinforce and validate the outputs generated by the traditional prediction models, increasing confidence in the results produced
7. Deliver these combined insights through an accessible platform that maximizes their practical usefulness for anyone seeking to make informed decisions about the market
8. Provide multiple ways for users to access and interact with these insights, rather than limiting them to a single, rigid interface

1.4 Final Result

1. The project succeeded in producing a fully integrated platform capable of covering an extensive range of the market, analyzing and generating predictions for over 15,000 individual stocks spread across more than 5 different stock exchanges worldwide.
2. Rather than discarding conventional analytical techniques, the finished system retained and strengthened them, using established methods as the foundation upon which additional predictive capability was layered.
3. A pipeline was established to continuously gather news content directly tied to measurable stock price movement, sourced from outlets relevant to financial and market activity.
4. This news content was run through a sentiment analysis process, producing a clear signal as to whether the tone and implications of recent coverage point toward an upward or downward movement in a given stock's price.
5. The sentiment-derived signal was not kept separate from the numerical prediction outputs; instead, the two were merged into one cohesive forecasting mechanism, so that every prediction reflects both angles simultaneously.
6. This merging process also serves a validating function: the sentiment layer acts as a cross-check against the model's traditional forecasts, giving users greater confidence that the predictions they see have been reinforced from more than one angle.
7. A complete, user-facing website was delivered, consolidating every layer of the project's output, the underlying analysis, the price forecasts, and the news-based sentiment findings into a single point of access.
8. Finally, a Power BI layer was incorporated into the finished system, providing an additional, more advanced analytical lens intended to sharpen and support the decision-making process for anyone relying on the platform.

2 Implementation Aspects

2.1 System Architecture

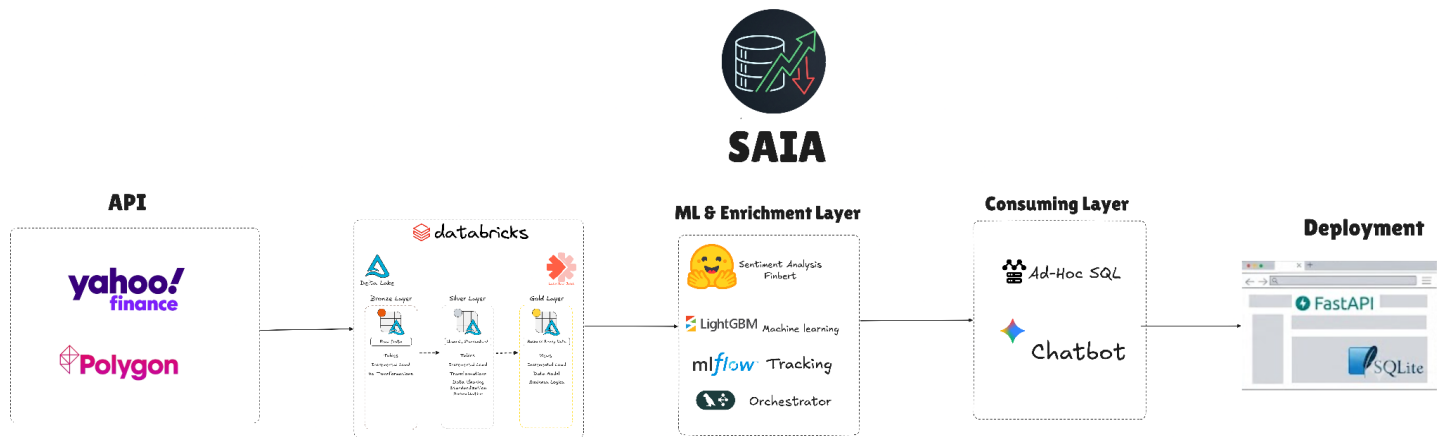


Figure 2.1: System Architecture

The overall architecture of the system is designed as a layered pipeline, moving data from its raw external sources all the way through to the interfaces where end users ultimately consume it, with each stage along the way responsible for a specific transformation or purpose.

The process begins at the API layer, where raw market and financial data is pulled in from external providers, specifically Yahoo Finance and Polygon.io. These two sources supply the foundational data on which the entire system is built, covering pricing information, company data, and other relevant market details across the thousands of stocks and multiple exchanges the platform is designed to support.

Once ingested, this data flows into Databricks, which forms the core processing engine of the architecture and is organized around a medallion structure of three progressive layers, all built on Delta Lake and orchestrated through Lakeflow Jobs. In the Bronze Layer, data is stored exactly as it arrives from the source raw, untransformed, and loaded incrementally, preserving a faithful record of the original API responses.

This raw data then moves into the Silver Layer, where it undergoes cleaning and standardization: transformations are applied, inconsistencies are resolved, and the data is normalized into a consistent, reliable structure, still maintained through incremental loading. From there, the data progresses into the Gold Layer, where it is shaped into business-ready form organized into views, enriched with business logic, and structured according to a proper data model, making it suitable for direct consumption by analytical and reporting tools.

From the Gold Layer, the refined data feeds into the Consuming Layer, where the platform's core analytical capabilities are applied. This includes Power BI, used to deliver advanced, visual analytics; a dedicated Sentiment Analysis component, responsible for interpreting the tone and implications of relevant news data; Machine Learning models, built to generate the stock predictions at the heart of the project; and support for Ad-Hoc SQL querying, allowing flexible, on-demand exploration of the underlying data.

Finally, the outputs generated across this consuming layer are surfaced to the end user through the website, which serves as the unified interface bringing together the analysis, predictions, and sentiment insights into a single, accessible platform.

This architecture will be discussed in greater depth in the coming chapters, with each layer and component explored in further technical detail but at a high level, it reflects the project's core approach: moving from raw, unstructured market data toward a refined, business-ready form, and ultimately toward actionable insight delivered directly to the user.

2.2 System Design

2.2.1 Online Analytical Processing (OLAP)

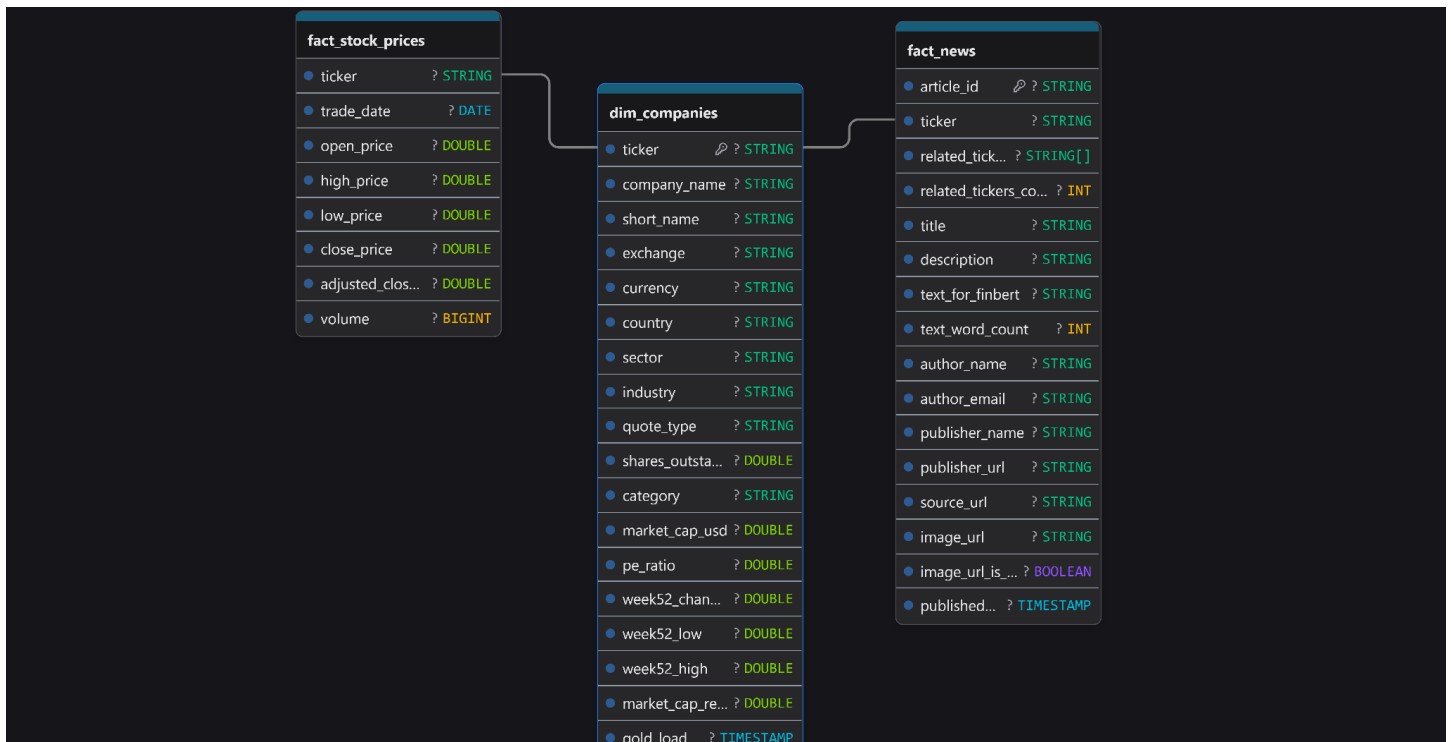


Figure 2.2: Star Schema

Fact Stock Prices : Historical daily stock prices from 2022 to the present.

Dim Companies : Master data for all companies and stock tickers.

Fact News : Financial news articles associated with stock tickers.

2.2.2 Machine Learning Digram

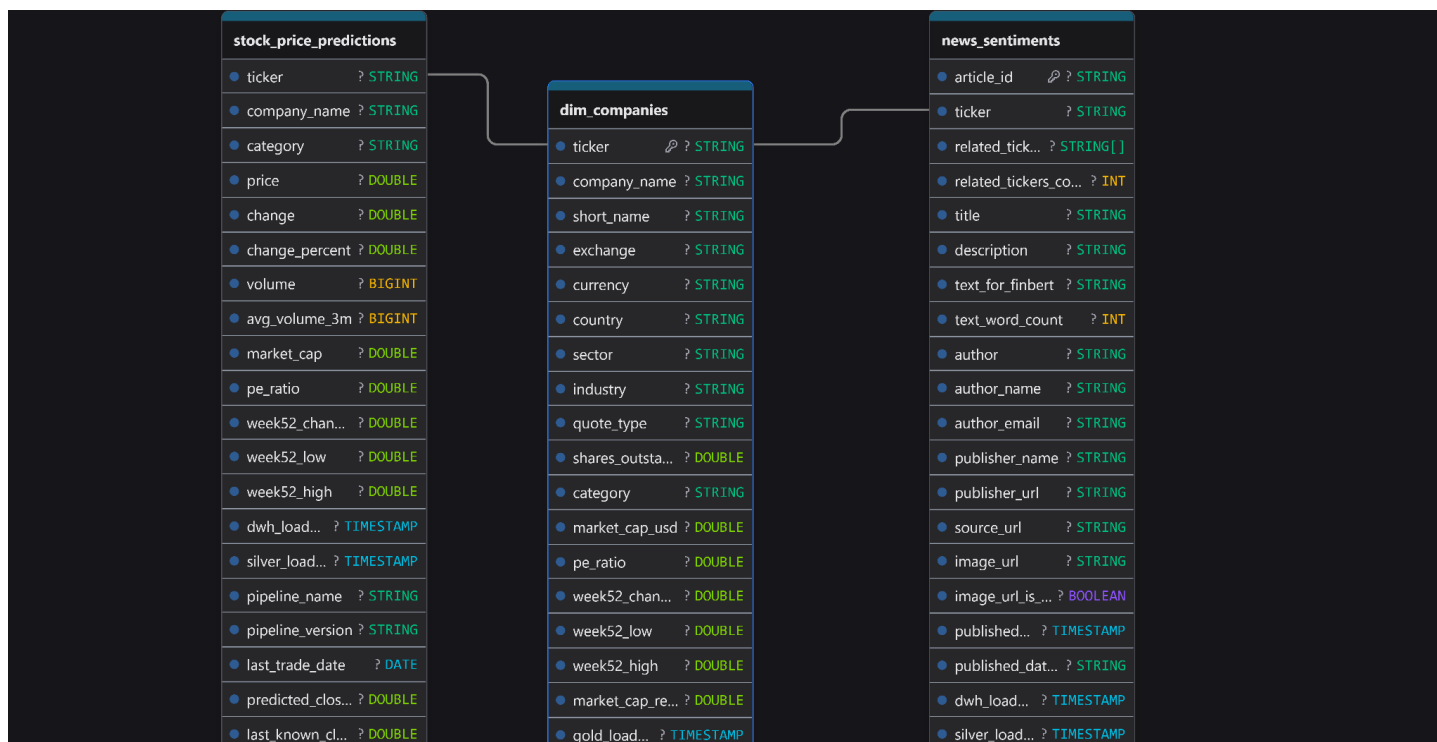


Figure 2.3: ML Digram

News Sentiments : Financial news sentiment scores generated using FinBERT
Stock Price Predictions : Machine learning predictions for future stock closing prices

2.3 Tools and Technologies

2.3.1 Data Engineering Technologies

- **Databricks** served as the main analytics and processing platform, enabling large-scale data transformation, integration, and analytical operations.
- **Delta Lake** was used to provide reliable and efficient data storage with support for data consistency, versioning, and optimized analytical workloads.

- **Databricks Asset Bundles** were used to manage, package, and deploy analytics resources across different environments, improving project organization and deployment consistency.
- **Python** was utilized for data extraction, transformation, automation, and machine learning development throughout the project.
- **Lakeflow Jobs** were used to automate and schedule data processing workflows within the analytics environment.
- **Apache Spark** was used as the distributed processing engine within Databricks to handle large-scale data transformation, aggregation, and analytical workloads efficiently.
- **PySpark** was used to perform large-scale data transformation, cleansing, and processing tasks within the Databricks environment.
- **Spark SQL** was utilized to query, manipulate, and analyze large datasets efficiently using SQL-based operations on Spark.
- **Git and GitHub** were used for **version control**, source code management, and collaborative development throughout the project lifecycle.

2.3.2 FullStack Web Technologies

- **Python:** The foundational language powering the entire backend architecture.
- **FastAPI:** A modern, high-performance, asynchronous Web framework used to build the RESTful APIs. It handles fast data streaming and automatically generates interactive Swagger documentation for seamless frontend integration.

- **Uvicorn:** The lightning-fast ASGI server utilized to run and serve the FastAPI application.
- **HTML5:** Used to build the semantic structure of the web platform, including the layout, Hero section, and interactive FAQ components.
- **CSS3 / Tailwind CSS:** Responsible for the responsive styling, modern layouts, and visual theme of the dashboard, ensuring a clean financial-tool look.
- **JavaScript (ES6+):** Handles client-side logic, dynamic UI rendering, component behavior, and client-state management directly in the user's browser.
- **SQLite:** A lightweight, embedded database dedicated to handling application-level state, such as user watchlists, chat sessions, and historical logs.
- **SQLAlchemy / Databases:** Used as the asynchronous ORM/query builder layer to securely interact with the SQLite database.

2.3.3 Machine learning & BI Technologies

- **Power BI & Power BI Service** was used to develop interactive dashboards and visualizations that support business intelligence and decision-making processes.
- **LightGBM** served as the core gradient boosting framework for predicting target stock close prices, selected for its high efficiency, fast training speed, and native handling of categorical features like company tickers.
- **Hyperopt** was used to perform automated hyperparameter tuning using the Tree-structured Parzen Estimator (TPE) algorithm to optimize parameters such as learning rate and number of leaves to minimize prediction error.
- **MLflow** served as the central platform for experiment tracking and model management, logging hyperparameters, tuning metrics, final evaluation results, and archiving the trained model artifacts.
- **Scikit-Learn** was used to provide statistical evaluation utilities, specifically computing performance metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) on the held-out test data.

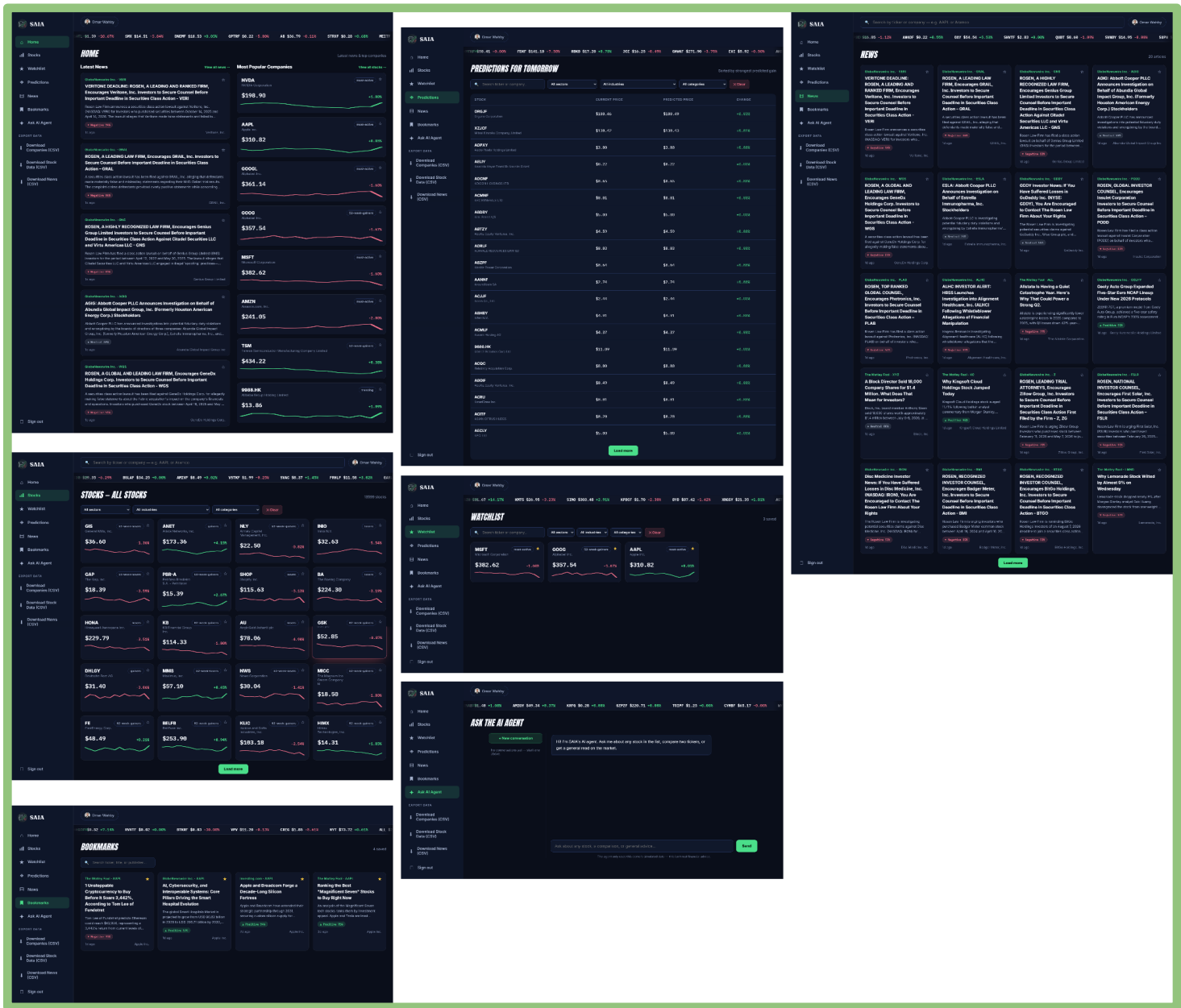
2.3.4 NLP

- **FinBERT (via Hugging Face Transformers)** served as the specialized Natural Language Processing model to analyze financial news texts and calculate sentiment scores across positive, negative, and neutral dimensions.
- **PyTorch** was used as the underlying deep learning framework to execute the FinBERT model pipelines, leveraging GPU acceleration and batched processing for optimized inference.
- **Apache Spark** was used to build high-performance distributed pipelines, utilizing native Spark window functions to compute scale-free lag, rolling average, and historical volatility features efficiently.
- **Delta Lake** served as the transactional storage layer across the pipeline stages, enabling idempotent database merges (**MERGE INTO**) to prevent duplicate article records and optimizing lookups via Z-Order indexing.

2.3.5 LLM & Agentic Chatbot Technologies

- **LangGraph** served as the core orchestration framework to construct a state-driven conversational agent, structuring execution via a directed graph that dynamically routes user queries between the model and an automated tool execution node.
- **Gemini (via LangChain Google GenAI)** served as the central intelligent engine, configured with a precise reasoning budget to ensure reliable multi-tool selection, prevent data hallucination, and generate natural responses in both English and Arabic.
- **SqliteSaver** was used to provide a persistent, transactional checkpointer that stores multi-turn conversation histories by session identifiers, maintaining durable chat states across application updates and service restarts.
- **LangChain Core Utilities** were used to manage messaging states, implement token-budget trimming algorithms to optimize context window efficiency, and dynamically append extracted cross-conversation user facts into the system instruction pipeline.

2.4 Prototype



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AB\$36.79-0.11%

STRXF\$0.28+0.68%

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Rosen Law Firm announces a securities class action lawsuit against Veritone, Inc. (NASDAQ: VERI) for investors who purchased securities between October 14, 2025 and April 14, 2026. The lawsuit alleges that Veritone made false statements and failed to...

▼ Negative 94%

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Veritone, Inc.

GlobeNewswire Inc. - GRAL

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ROSEN, A LEADING LAW FIRM, Encourages GRAIL, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action - GRAL

A securities class action lawsuit has been filed against GRAIL, Inc. alleging that defendants made materially false and misleading statements regarding their NHS-Galleri trial results. The complaint claims defendants provided overly positive statements while concealing...

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GRAIL, Inc.

GlobeNewswire Inc. - GNS

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ROSEN, A HIGHLY RECOGNIZED LAW FIRM, Encourages Genius Group Limited Investors to Secure Counsel Before Important Deadline in Securities Class Action Against Citadel Securities LLC and Virtu Americas LLC - GNS

Rosen Law Firm has filed a class action lawsuit on behalf of Genius Group Limited (GNS) investors for the period between April 12, 2022 and May 30, 2025. The lawsuit alleges that Citadel Securities LLC and Virtu Americas LLC engaged in illegal 'spoofing' practices—...

▼ Negative 83%

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Genius Group Limited

GlobeNewswire Inc. - AGIG

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AGIG: Abbott Cooper PLLC Announces Investigation on Behalf of Abundia Global Impact Group, Inc. (Formerly Houston American Energy Corp.) Stockholders

Abbott Cooper PLLC has announced investigations into potential fiduciary duty violations and wrongdoing by the boards of directors of three companies: Abundia Global Impact Group, Inc. (formerly Houston American Energy Corp.), Estrella Immunopharma, Inc., and...

● Neutral 69%

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Abundia Global Impact Group Inc

GlobeNewswire Inc. - WGS

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ROSEN, A GLOBAL AND LEADING LAW FIRM, Encourages GeneDx Holdings Corp. Investors to Secure Counsel Before Important Deadline in Securities Class Action - WGS

A securities class action lawsuit has been filed against GeneDx Holdings Corp. for allegedly making false statements about the Fabric acquisition's impact on the company's financials and operations. Investors who purchased GeneDx stock between April 16, 2025 and May ...

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GeneDx Holdings Corp.

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NVDA

NVIDIA Corporation

most-active ☆

\$198.90

+1.00%

AAPL

Apple Inc.

most-active ☆

\$310.82

+0.05%

GOOGL

Alphabet Inc.

most-active ☆

\$361.14

-1.60%

GOOG

Alphabet Inc.

52-week-gainers ☆

\$357.54

-1.67%

MSFT

Microsoft Corporation

most-active ☆

\$382.62

-1.60%

AMZN

Amazon.com, Inc.

most-active ☆

\$241.05

-2.00%

TSM

Taiwan Semiconductor Manufacturing Company Limited

52-week-gainers ☆

\$434.22

+0.38%

9988.HK

Alibaba Group Holding Limited

trending ☆

\$13.86

+1.09%

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DR \$29.35 -6.29%

BSLAF \$34.25 +0.00%

AMIVF \$8.49 +0.02%

VSTKF \$1.99 -0.25%

SVAC \$8.37 +1.45%

FRHLF \$11.58 +3.02%

GARL

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GISGeneral Mills, Inc.

52-week-losers

\$36.60

-1.36%

ANETArista Networks, Inc.

gainers

\$173.36

+4.15%

NLYAnnaly Capital Management, Inc.

52-week-gainers

\$22.50

-0.82%

INIOInnio N.V.

losers

\$32.63

-5.54%

GAPThe Gap, Inc.

52-week-losers

\$18.39

-3.59%

PBR-APetróleo Brasileiro S.A. - Petrobras

52-week-gainers

\$15.39

+2.67%

SHOPShopify Inc.

losers

\$115.63

-5.13%

BAThe Boeing Company

losers

\$224.30

-3.19%

HONAHoneywell Aerospace Inc.

losers

\$229.79

-3.51%

KBKB Financial Group Inc.

52-week-gainers

\$114.33

-1.00%

AUAngloGold Ashanti plc

losers

\$78.06

-4.90%

GSK

52-week-gainers

\$52.85

-0.87%

DHLGYDeutsche Post AG

gainers

\$31.40

-3.06%

MMSMaximus, Inc.

52-week-losers

\$57.10

+0.45%

NWSNews Corporation

52-week-losers

\$30.04

-1.41%

MICCThe Magnum Ice Cream Company N.

52-week-gainers

\$18.50

-1.80%

FEFirstEnergy Corp.

52-week-gainers

\$48.49

+0.21%

BELFBBel Fuse Inc.

52-week-gainers

\$253.90

+0.94%

KLICKulicke and Soffa Industries, Inc.

52-week-gainers

\$103.18

-2.54%

HIMXHimax Technologies, Inc.

52-week-gainers

\$14.31

+1.85%

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GeneDx Holdings Corp.

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Estrella Immunopharma, Inc.

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GDDY Investor News: If You Have Suffered Losses in GoDaddy Inc. (NYSE: GDDY), You Are Encouraged to Contact The Rosen Law Firm About Your Rights

The Rosen Law Firm is investigating potential securities claims against GoDaddy Inc., Wise Group plc, and...

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GoDaddy Inc.

GlobeNewswire Inc. - PODO ☆

ROSEN, GLOBAL INVESTOR COUNSEL, Encourages Insulet Corporation Investors to Secure Counsel Before Important Deadline in Securities Class Action - PODO

Rosen Law Firm has filed a class action lawsuit against Insulet Corporation (PODD) on behalf of investors who...

▼ Negative 83%

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Insulet Corporation

GlobeNewswire Inc. - PLAB ☆

ROSEN, TOP RANKED GLOBAL COUNSEL, Encourages Photonics, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action - PLAB

Rosen Law Firm has filed a class action lawsuit against Photonics, Inc. (NASDAQ: PLAB) on behalf of investors who...

▼ Negative 92%

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Photonics, Inc.

GlobeNewswire Inc. - ALHC ☆

ALHC INVESTOR ALERT: HBSS Launches Investigation into Alignment Healthcare, Inc. (ALHC) Following Whistleblower Allegations of Financial Manipulation

Hagens Berman is investigating Alignment Healthcare (ALHC) following whistleblower allegations that the...

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Alignment Healthcare, Inc.

The Motley Fool - ALL ☆

Allstate Is Having a Quiet Catastrophe Year. Here's Why That Could Power a Strong Q2.

Allstate is experiencing significantly lower catastrophe losses in 2026 compared to 2025, with Q1 losses down 43% year-...

▼ Negative 77%

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The Allstate Corporation

GlobeNewswire Inc. - GELHY ☆

Geely Auto Group Expanded Five-Star Euro NCAP Lineup Under New 2026 Protocols

ZEEKR 7GT, a premium model from Geely Auto Group, achieved a five-star safety rating in Euro NCAP's 2026 assessment...

▲ Positive 92%

1d ago

Geely Automobile Holdings Limited

The Motley Fool - XYZ ☆

A Block Director Sold 18,000 Company Shares for \$1.4 Million. What Does That Mean for Investors?

Block, Inc. board member Anthony Eisen sold 18,000 shares worth approximately \$1.4 million between July 6-8, 2026, at ...

● Neutral 80%

1d ago

Block, Inc.

The Motley Fool - KC ☆

Why Kingsoft Cloud Holdings Stock Jumped Today

Kingsoft Cloud Holdings stock surged 11.71% following bullish analyst commentary from Morgan Stanley....

▲ Positive 95%

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Kingsoft Cloud Holdings Limited

GlobeNewswire Inc. - Z ☆

ROSEN, LEADING TRIAL ATTORNEYS, Encourages Zillow Group, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action First Filed by the Firm - Z, ZG

Rosen Law Firm is urging Zillow Group investors who purchased stock between February 11, 2025 and May 7, 2026 to jo...

▼ Negative 71%

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Zillow Group, Inc.

GlobeNewswire Inc. - FSLR ☆

ROSEN, NATIONAL INVESTOR COUNSEL, Encourages First Solar, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action - FSLR

Rosen Law Firm is urging First Solar, Inc. (FSLR) investors who purchased securities between February 26, 2025...

▼ Negative 91%

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First Solar, Inc.

GlobeNewswire Inc. - IRON ☆

Disc Medicine Investor News: If You Have Suffered Losses in Disc Medicine, Inc. (NASDAQ: IRON), You Are Encouraged to Contact The Rosen Law Firm About Your Rights

The Rosen Law Firm is investigating potential securities claims against Disc Medicine, Inc. (NASDAQ: IRON) for...

▼ Negative 93%

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Disc Medicine, Inc.

GlobeNewswire Inc. - BMI ☆

ROSEN, RECOGNIZED INVESTOR COUNSEL, Encourages Badger Meter, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action - BMI

Rosen Law Firm is urging investors who purchased Badger Meter common stock between April 18, 2024 and April 16, 20...

▼ Negative 85%

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Badger Meter, Inc.

GlobeNewswire Inc. - BTGO ☆

ROSEN, RECOGNIZED INVESTOR COUNSEL, Encourages BitGo Holdings, Inc. Investors to Secure Counsel Before Important Deadline in Securities Class Action - BTGO

Rosen Law Firm is reminding BitGo Holdings investors of an August 7, 2026 deadline to join a securities class action...

▼ Negative 93%

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BitGo Holdings, Inc.

The Motley Fool - LMND ☆

Why Lemonade Stock Wilted by Almost 9% on Wednesday

Lemonade stock dropped nearly 9% after Morgan Stanley analyst Bob Huang downgraded the stock from overweight ...

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FDXF\$141.10-7.58%

RKBK\$17.20+0.70%

JCE\$16.25-0.49%

GMMKF\$271.90-3.75%

EH1\$5.92-0.50%

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STOCK	CURRENT PRICE	PREDICTED PRICE	CHANGE
ORGJF Organo Corporation	\$100.46	\$100.49	+0.03%
XZJCF Mitsui Kinzoku Company, Limited	\$130.42	\$130.43	+0.01%
ADPXY Audio Pixels Holdings Limited	\$3.00	\$3.00	+0.00%
AEIY Anadolu Hayat Emeklilik Anonim Sirketi	\$8.22	\$8.22	+0.00%
ADCNF ADICON HOLDINGS LTD	\$0.65	\$0.65	+0.00%
ACMNF ARC MINERALS LTD	\$0.01	\$0.01	+0.00%
ABDBY Alm. Brand A/S	\$5.00	\$5.00	+0.00%
ABTZY Aboitiz Equity Ventures, Inc.	\$4.59	\$4.59	+0.00%
ADRLF ADAVALE RESOURCES LIMITED	\$0.03	\$0.03	+0.00%
ABZPF Aboitiz Power Corporation	\$0.64	\$0.64	+0.00%
AANNF Aroundtown SA	\$2.74	\$2.74	+0.00%
ACJJF Acom Co., Ltd.	\$2.44	\$2.44	+0.00%
ABHBY Alfen N.V.	\$4.91	\$4.91	+0.00%
ACMLF Ascom Holding AG	\$4.37	\$4.37	+0.00%
9880.HK Ubtech Robotics Corp Ltd	\$11.09	\$11.09	+0.00%
ACQC Relativity Acquisition Corp.	\$8.00	\$8.00	+0.00%
ABOIF Aboitiz Equity Ventures, Inc.	\$0.49	\$0.49	+0.00%
ACRU AmeriCrew Inc.	\$0.01	\$0.01	+0.00%
ACITF ASIAN CITRUS HLDGS	\$0.20	\$0.20	+0.00%
AECLY AECI Ltd	\$5.00	\$5.00	+0.00%

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NABI \$1.40 +1.08%AMIGY \$49.34 +0.37%KRFG \$0.28 +0.00%GZPZF \$220.71 +0.00%TOIPF \$1.25 +0.00%CYMBF \$65.17 -0.00%WV

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+ New conversation

No conversations yet — start one above.

Hi! I'm SAIA's AI agent. Ask me about any stock in the list, compare two tickers, or get a general read on the market.

Ask about any stock, a comparison, or general advice...

Send

The agent only uses this demo's simulated data — this isn't real financial advice.

Data Engineering

19

3.1 Data Sources

The primary data sources for the Stock and Arrows Investing App are Yahoo Finance and Polygon.io, two external providers that together supply the full range of data required for stock analysis and prediction. Yahoo Finance serves as the main source for stock market data, providing pricing information, company details, and other financial metrics across the wide universe of stocks covered by the platform, more than 15,000 stocks spread across more than 5 stock exchanges. Polygon.io, on the other hand, is used primarily as the source for news data, supplying the articles and financial news content that feed directly into the sentiment analysis component of the system.

In terms of scale, the platform draws on a substantial volume of data to support its analysis and prediction capabilities. The historical stock pricing data collected spans back to 2022 and covers approximately 17,000 individual stocks, amounting to roughly 16 million rows of historical data in total. Alongside this, the news data gathered from Polygon.io comprises around 500,000 individual news articles, forming the basis on which sentiment analysis is performed across the covered stocks.

Given the scale of data involved spanning thousands of individual stocks, multiple exchanges, and continuously updating news content a careful assessment process was carried out to determine which specific data points from each source held real analytical value for the platform's goals. Rather than pulling every available field indiscriminately, the focus was placed on the data most relevant to the project's core objectives, including:

- Stock pricing data (open, high, low, close, volume)
- Company and exchange information
- Currency and exchange rate data, to normalize pricing across different markets
- Financial news content relevant to specific companies and stocks

Together, these datasets form the foundation on which all subsequent processing, analysis, sentiment evaluation, and prediction activities within the platform are built.

3.2 Extract Layer

To bring data in from these external sources, a dedicated ingestion layer was developed to connect to both the Yahoo Finance and Polygon.io APIs, retrieve the relevant data, and load it into the analytics environment for further processing. This ingestion layer was designed to handle the scale and nature of financial market data specifically accounting for the large number of stocks being tracked, the need for regular and reliable updates as prices and news continuously change, and the different structures and formats returned by each of the two data sources.

The extract process was built to operate consistently across both data sources, ensuring that pricing data from Yahoo Finance and news data from Polygon.io are each collected in a reliable, automated, and structured manner, forming the entry point into the platform's broader data pipeline.

3.3 Medallion Data Architecture

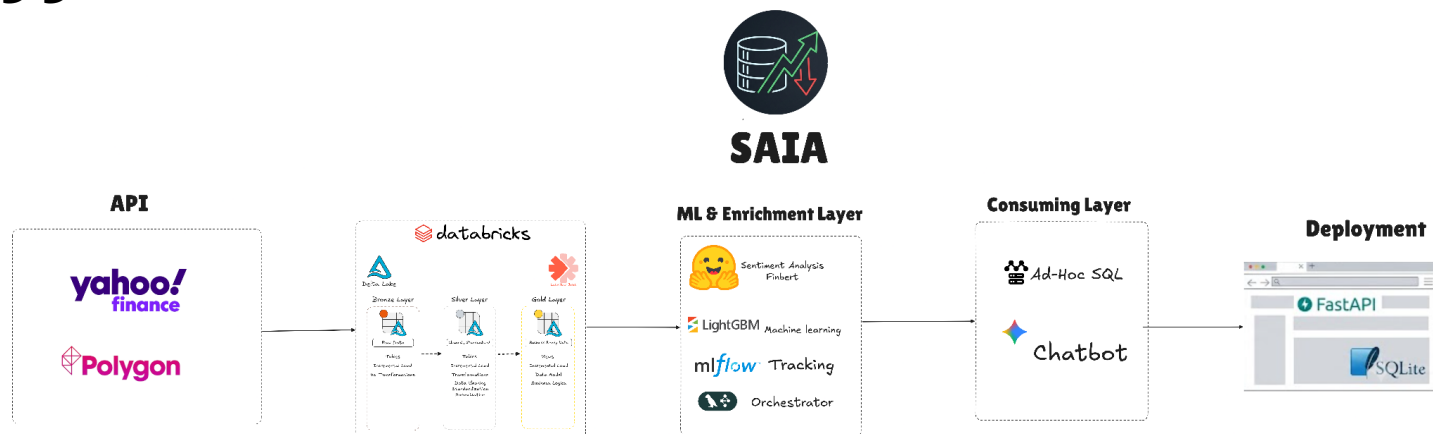


Figure 5.1: Medallion Data Architecture

Figure 5.1 illustrates the Medallion Data Architecture adopted in the Stock and Arrows Investing App. The architecture is designed to transform raw market and financial data extracted from external providers Yahoo Finance and Polygon.io into business-ready analytical datasets. Data progresses through a series of refinement layers, where it is ingested, standardized, and modeled before being delivered to the platform's reporting, analytics, prediction, and decision-support components. This layered approach improves data quality, scalability, and maintainability, while providing a reliable foundation for large-scale stock analysis and prediction across thousands of stocks and multiple exchanges.

3.4 Bronze Layer

The Bronze Layer serves as the raw data ingestion layer within the Stock and Arrows Investing App. Data extracted from the external providers Yahoo Finance and Polygon.io is stored in Delta Lake tables while preserving the original source structure, including field names and formats as returned by each provider's API.

The initial data load is performed as a full load, capturing the historical stock pricing and news data described earlier, while subsequent loads are executed incrementally, using a load timestamp watermark to identify and capture only newly created or updated records since the last run.

This approach enables the platform to efficiently keep pace with continuously updating stock prices and newly published news articles, while minimizing unnecessary data movement and processing overhead.

To maintain complete traceability, metadata columns such as `dwh_source_db` are added during ingestion, allowing every record to be traced back to the specific load that brought it into the platform. No business transformations, validations, or data cleansing operations are applied at this stage.

The primary objective of the Bronze Layer is to maintain an accurate and reliable copy of the data exactly as it was received from Yahoo Finance and Polygon.io, providing a consistent and trustworthy foundation for the downstream processing carried out in the Silver and Gold layers.

Any new stock prices or news articles retrieved from these sources are automatically captured in the Bronze Layer through the incremental loading mechanism, ensuring the platform stays continuously up to date with the latest available market and news data.

3.5 Silver Layer

The Silver Layer serves as the data preparation and standardization layer within the Stock and Arrows Investing App. Data is loaded incrementally from the Bronze Layer using the same change-tracking mechanism applied at the ingestion stage, ensuring that new and updated stock prices and news records are continuously propagated through the pipeline.

At this stage, data is transformed into a more structured and analytics-ready format. Key processing activities include data type standardization, naming convention alignment, light data cleansing, and normalization. Source attributes stored as strings in the Bronze Layer are converted to their appropriate data types, while column names are standardized to maintain consistency across datasets. In addition, currency conversion is applied at this stage, with stock prices originally recorded in local currencies converted into a standardized USD value, ensuring consistency across stocks listed on different exchanges around the world.

One of the features that distinguishes this project's Silver Layer is the introduction of a dedicated `silver_rejected` schema, used to systematically identify and isolate low-quality or unreliable data before it can propagate further into the pipeline. Through this mechanism, stocks and companies whose data was found to be invalid, missing, or misleading for example, records containing zero values or other clearly erroneous entries were flagged and excluded from the main Silver dataset rather than being carried forward into later stages. This data quality filtering process proved to be a valuable outcome of the project, resulting in the removal of 4,000 companies and more than 1 million rows of

unreliable stock data, ensuring that only trustworthy, validated data continues on to the Gold Layer.

The resulting datasets provide a clean, consistent, and well-structured foundation for the Gold Layer, where the business-oriented analytical models including the prediction and sentiment analysis components are developed.

3.6 Gold Layer

The Gold Layer represents the final, business-ready stage of the pipeline, where data originating from the Silver Layer is shaped into a structure optimized for analysis, reporting, and downstream consumption by the platform's prediction and sentiment analysis components. Rather than working with raw or intermediate structures, this layer is organized around a Star Schema, designed specifically to support efficient querying and analysis across both stock price data and news content.

At the center of this schema sits a single shared dimension table, `dim_companies`, which holds descriptive attributes for each company covered by the platform — including its ticker, company name, exchange, sector, industry, currency, and other company-level metrics such as market capitalization and 52-week price ranges. This dimension table serves as the common reference point connecting the platform's two core fact tables.

The first of these, `fact_stock_prices`, holds the core historical pricing data for each stock, including open, high, low, close, and adjusted close prices, along with daily trading volume, forming the foundation on which the platform's prediction models are built. The second, `fact_news`, holds the collected news articles relevant to one or more tracked companies, along with associated metadata such as publisher and author details, source information, and a pre-processed text field prepared specifically for sentiment analysis using FinBERT.

By linking both fact tables back to the same `dim_companies` dimension, the Gold Layer enables unified analysis and querying across both stock price movement and news activity, using the company as the shared point of

reference. In total, this stage of the pipeline currently consists of three Gold-level datasets, forming a clean, business-ready foundation on which the platform's Power BI dashboards, sentiment analysis, and machine learning prediction models are all ultimately built

3.7 Data Pipeline Orchestration

To ensure fully automated data processing, Stock and Arrows implements an orchestration framework using Databricks Workflows and Databricks Asset Bundles. The orchestration layer manages the execution sequence of all data engineering tasks, ensuring that dependencies between pipeline stages are respected and that data flows consistently across the Medallion Architecture.

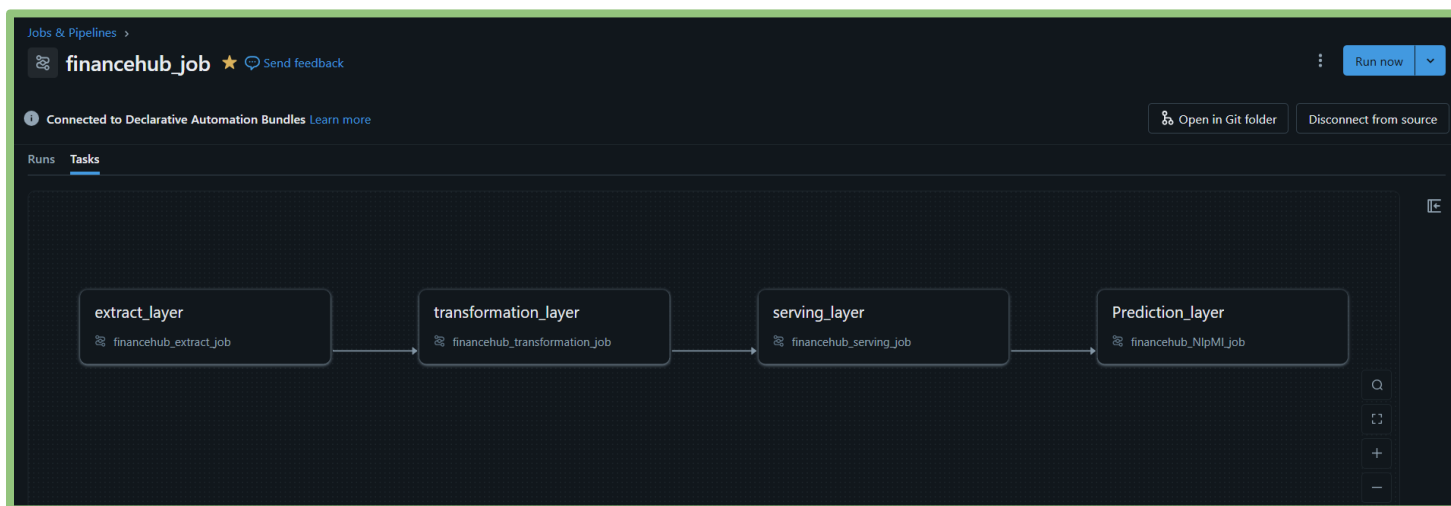
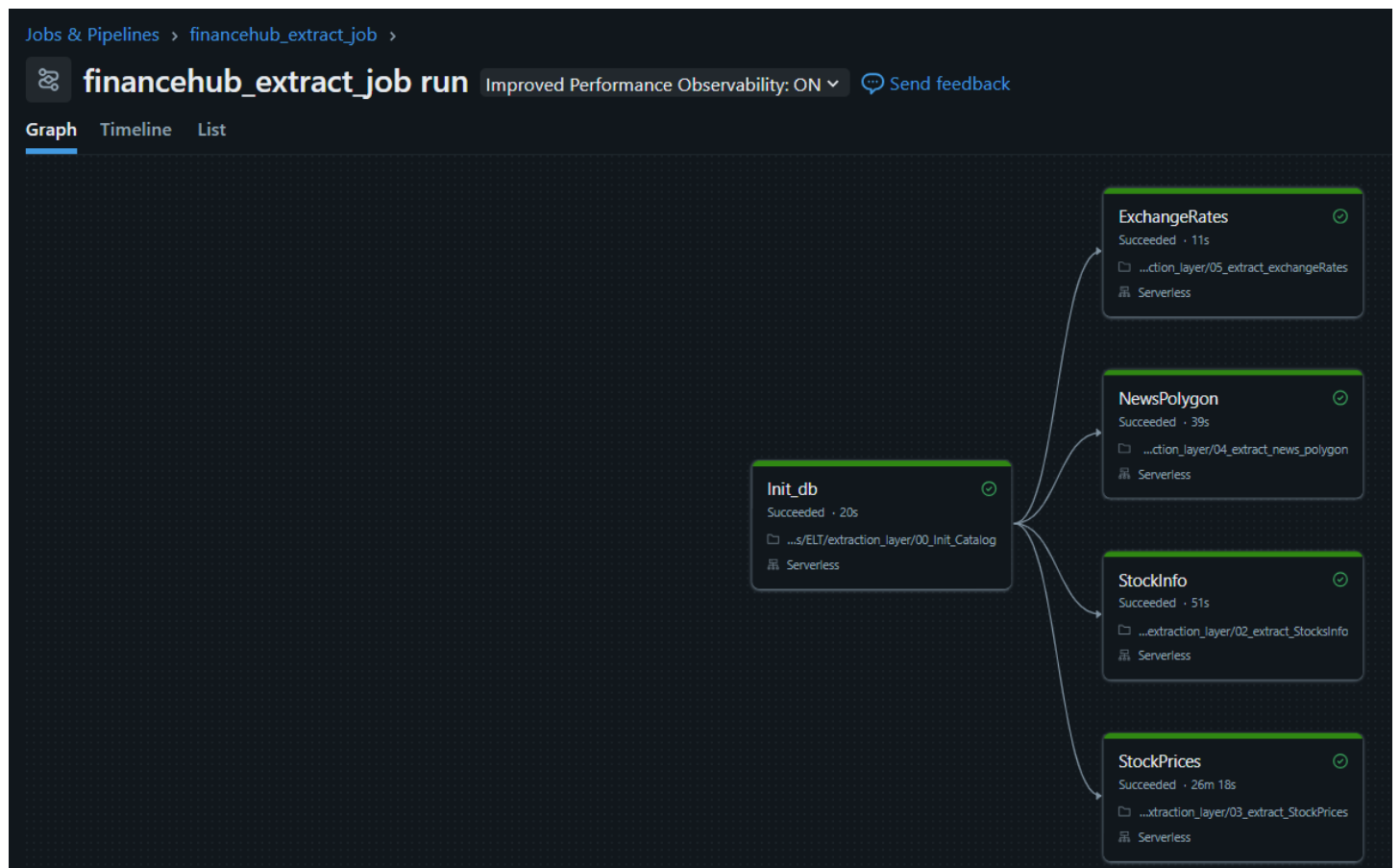


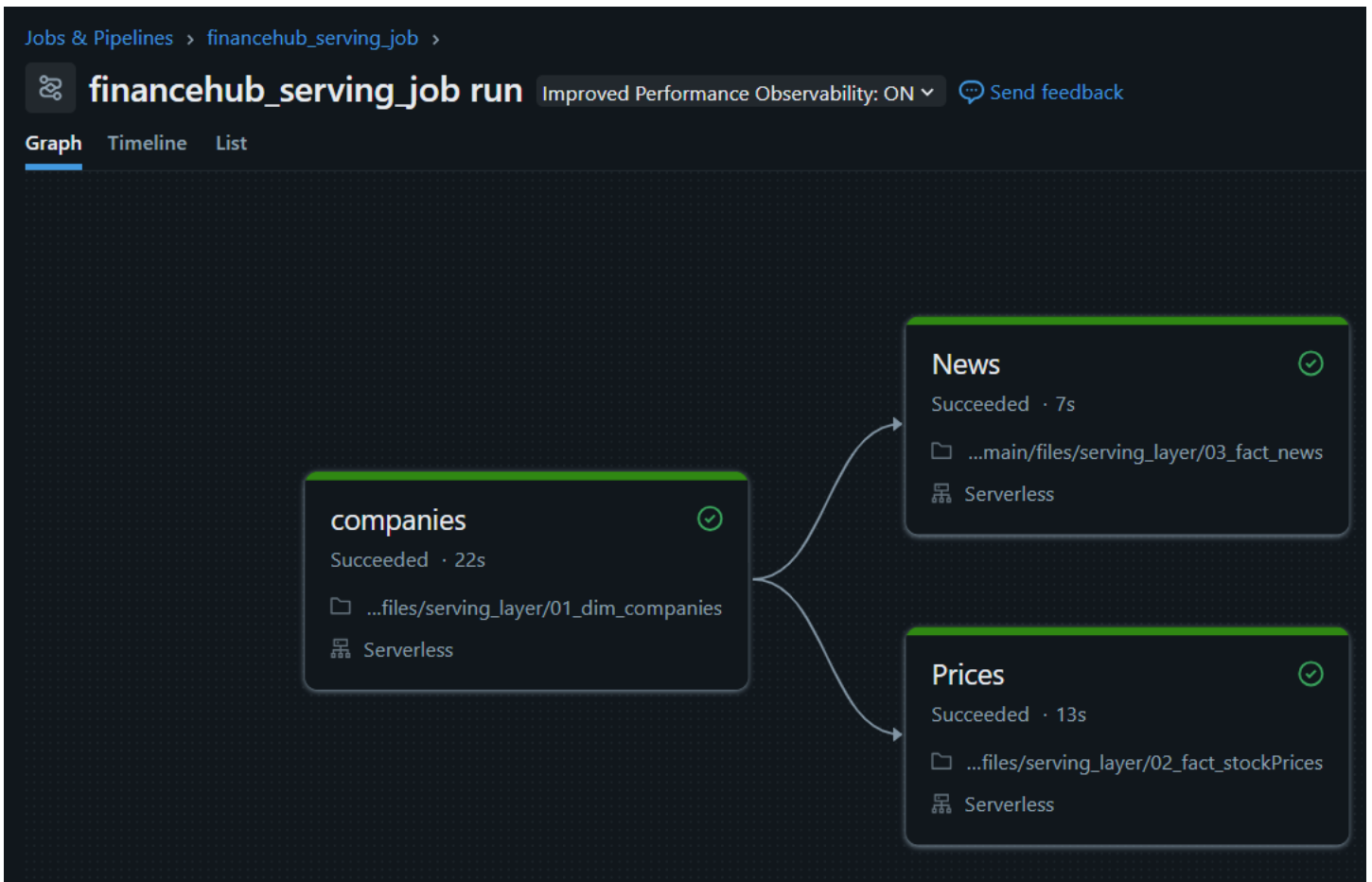
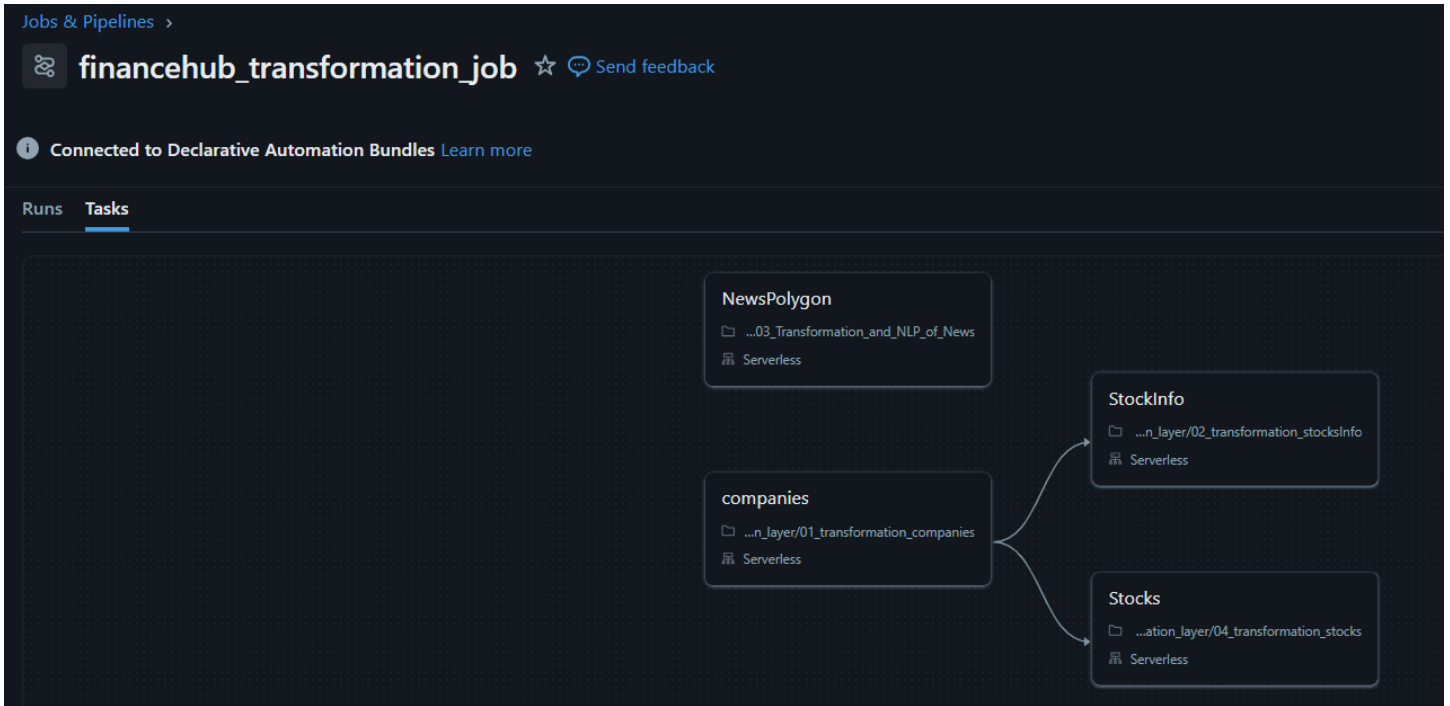
Figure 3.2: Master Pipeline

Figure 3.2 illustrates the master orchestration workflow responsible for coordinating all data engineering processes within Stock and Arrows.

The Master Job orchestrates the execution of all pipeline stages through predefined task dependencies. The pipeline is structured into three sequential sub-jobs: the extraction job, which retrieves market data and financial news from external APIs; the transformation job, which processes and standardizes the ingested data through the Bronze and Silver layers; and the serving job, which constructs the Gold Layer analytical tables and views consumed by downstream applications. Each sub-job is triggered only after the successful completion of its

predecessor, ensuring data consistency and preventing downstream processes from operating on incomplete datasets.





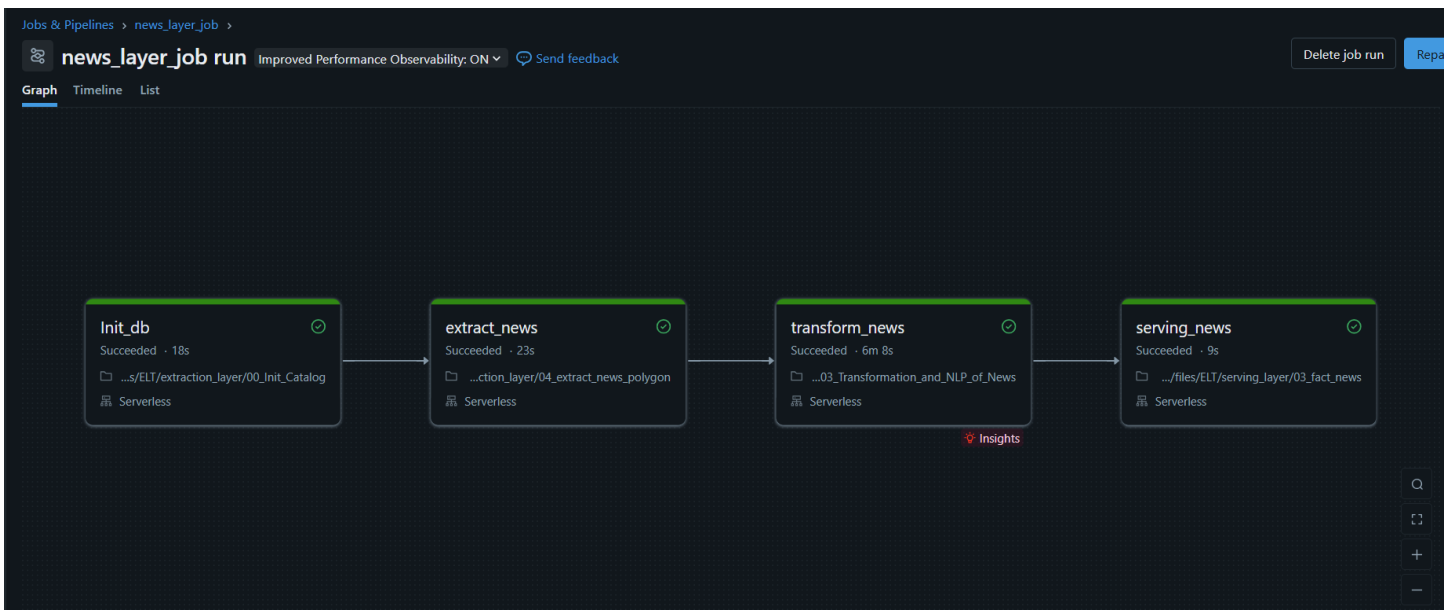
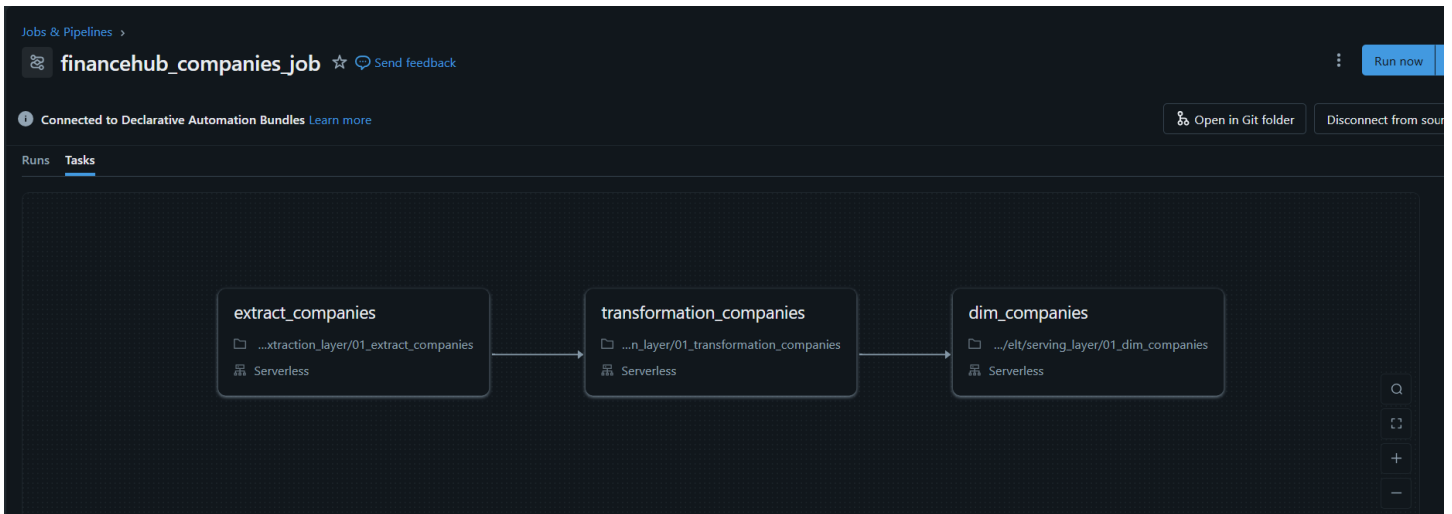


Figure 3.3: Data Pipeline Orchestration Components

Figure 3.3 presents the orchestration components implemented in Stock and Arrows. Databricks Workflows were used to automate pipeline execution and schedule daily market data refreshes aligned with exchange operating hours, while Databricks Asset Bundles provided deployment and version control capabilities through a Git-based workflow.

Databricks Asset Bundles (DAB)

Databricks Asset Bundles were used to manage and deploy all data engineering assets through a Git-based development workflow. This approach enables version control, reproducible deployments, and centralized management of notebooks, workflows, and configuration files across the project environment. The bundle definition organizes all pipeline resources — including extraction notebooks, transformation jobs, and Gold Layer serving logic — into a structured and maintainable project layout that supports consistent deployment across development and production environments.

Incremental Processing

To improve efficiency and reduce processing time, incremental loading was implemented using a high-watermark pattern based on the **dwh_loaded_at** timestamp field. This approach allows only newly ingested or modified records to be propagated through the Silver and Gold layers during each pipeline execution, minimizing compute overhead and reducing end-to-end refresh latency. For stock price data, incremental filtering is applied at the exchange and symbol level, ensuring that only records introduced since the last successful pipeline run are reprocessed downstream.

Exchange Rate Snapshot Management

In addition to market data, the pipeline incorporates a dedicated daily snapshot mechanism for currency exchange rates sourced from the open.er-api.com API. Each pipeline execution captures a point-in-time exchange rate record for all supported currency pairs, enabling accurate USD price conversion for OHLC stock data across multiple international exchanges. Historical snapshots are preserved in the Gold Layer through a MERGE-based upsert pattern, ensuring that rate records remain traceable and auditable across time.

3.8 Power BI



Figure 3.4: Power BI Dashboard

Power BI Stock Analysis Dashboard

The figure presents the main stock analysis report built in Power BI as part of the *Stock and Arrows* project. The report is designed to replicate the look and feel of professional financial platforms (such as Yahoo Finance or Google Finance) using only native Power BI visuals, without relying on any third-party or custom visuals.

The dashboard is composed of the following key components:

Header Section: Displays the selected company name, the current stock price, and two dynamic variation cards showing the percentage and absolute price change over the selected time period, along with the change over the past year. These values are color-coded in green or red depending on whether the movement is positive or negative, giving the user an immediate visual read on stock performance.

Metadata Bar: Shows contextual information such as the date of the last available closing price, the trading currency, the exchange the stock is listed on, the country, sector/industry classification, and a market activity tag (e.g., "most-active").

Candlestick Chart: The central visual of the report, showing the daily Open, High, Low, and Close (OHLC) prices for the selected stock. Green candles indicate bullish days (closing price higher than opening price), while red candles indicate bearish days. A dotted moving-average line is overlaid on the chart to help visualize the underlying price trend, independent of daily volatility.

Time Period and Moving Average Controls: Positioned in the upper-right corner, these controls allow the user to adjust the moving-average window (20, 30, 40, or 50 days) and to filter the chart to a specific time range (1 day, 5 days, 1 month, 3 months, 6 months, year-to-date, 1 year, 2 years, or 5 years), enabling both short-term and long-term trend analysis.

Volume Chart: Displayed beneath the candlestick chart, this bar chart shows the daily trading volume, with bars colored green or red to match the direction of that day's price movement, allowing users to correlate trading activity with price trends.

Overall, this report provides an interactive, single-view financial dashboard where analysts and investors can explore historical stock behavior and short-term momentum for any company within the platform's coverage, entirely through native Power BI functionality.

4 AI Applications

4.1 Overview

This chapter documents the AI-driven components of the platform's intelligence layer — the parts of the system that go beyond descriptive dashboards and actively interpret, forecast, or converse. Three components are covered: a transformer-based news sentiment engine (FinBERT), a gradient-boosted price prediction model (LightGBM), and SAIA, a LangGraph-orchestrated conversational agent backed by Gemini. Together they turn raw ingested market data into scored, forecasted, and conversationally accessible intelligence.

All three components sit on the same Databricks Lakehouse foundation used elsewhere in the platform (bronze → silver → gold), and all three are designed to be idempotent and safely re-runnable: incremental loads use watermark-based discovery, writes use Delta MERGE or atomic overwrite, and temporary/scratch tables are cleaned up even on failure.

4.2 Data Processing and Feature Engineering

Sentiment cleaning

Before scoring, articles are cleaned with a small set of reusable transformations: blank-string-to-NULL normalization, whitespace/newline collapsing, related-ticker parsing into a deduplicated upper-cased array, URL validity checks, and author-name/email extraction via regular expressions. Because `dwh_loaded_at` appears in two formats across the backfill and the live incremental loader, both formats are attempted with `try_to_timestamp` and coalesced, rather than assuming a single fixed format. A stable `article_id` is derived by hashing the source URL (or, when absent, the title plus published date) with SHA-256, giving every article a deterministic surrogate key independent of ingestion order.

Sentiment-to-price alignment

News is not published only on trading days. To avoid silently dropping weekend or holiday news, each article's publish date is forward-mapped to the next actual trading date using a calendar-spine and forward-fill technique, rather than a direct date-equality join. Sentiment scores are then averaged per (ticker, trading date) before being joined onto the price series, so a Friday-evening headline still influences Monday's feature set instead of vanishing.

Price feature engineering

All price-prediction features are scale-free by construction, so the same global model behaves consistently for a \$2 stock and a \$2,000 stock: price and volume lags are expressed as log-returns over 1, 2, 3, 5 and 10-day horizons; rolling features compare today's value to its own 5- and 10-day rolling mean; and a 10-day rolling standard deviation of daily percentage change captures short-term volatility. In version 4 of the pipeline, this feature computation was moved from a per-ticker Pandas groupby/transform (a severe bottleneck across roughly 13,550 ticker groups) to native, distributed Spark window functions, cutting the step from several minutes to seconds without changing any feature definition.

4.3 AI Component Descriptions

4.3.1 News Sentiment Analysis Engine (FinBERT)

Each cleaned article's text is truncated to a fixed character budget and scored in batches by ProsusAI/finbert, a transformer model fine-tuned for financial-text sentiment. The model produces a positive, negative, and neutral probability for every article, which are merged into silver.news_sentiments with a Delta MERGE keyed on (article_id, ticker) — matched rows are updated, unmatched rows are inserted, so the job is safe to re-run without creating duplicates. The table is periodically OPTIMIZED and Z-ORDERED by (ticker, published_at) to keep ticker- and date-scoped reads fast as the table grows.

4.3.2 Stock Price Prediction Model (LightGBM)

Rather than training one model per ticker, the platform trains a single global LightGBM regressor across all tracked companies, with ticker encoded as a categorical feature. This lets the model share learned patterns across similar stocks while still specializing per ticker through the categorical split. The target is the log-return of next-day adjusted close price (not raw close), which keeps the target free of mechanical distortions from stock splits or dividend events.

Hyperparameter search uses Hyperopt's sequential Trials rather than SparkTrials: LightGBM's Dataset object holds C-level handles that are not safely picklable across worker processes, and LightGBM already parallelizes internally via threads, so distributing trials across executors would either break or simply create resource contention. To keep the search affordable at scale, each trial trains against a fixed random sample of tickers rather than the full dataset — the sample changes which companies are seen, never which dates are, so no trial is ever exposed to data beyond the training cutoff. Once the best hyperparameters are found, the final model is retrained on the complete dataset.

Companies with too little trading history are explicitly skipped at prediction time rather than fed fabricated feature values, and are reported separately with a NULL prediction and a last-known price only.

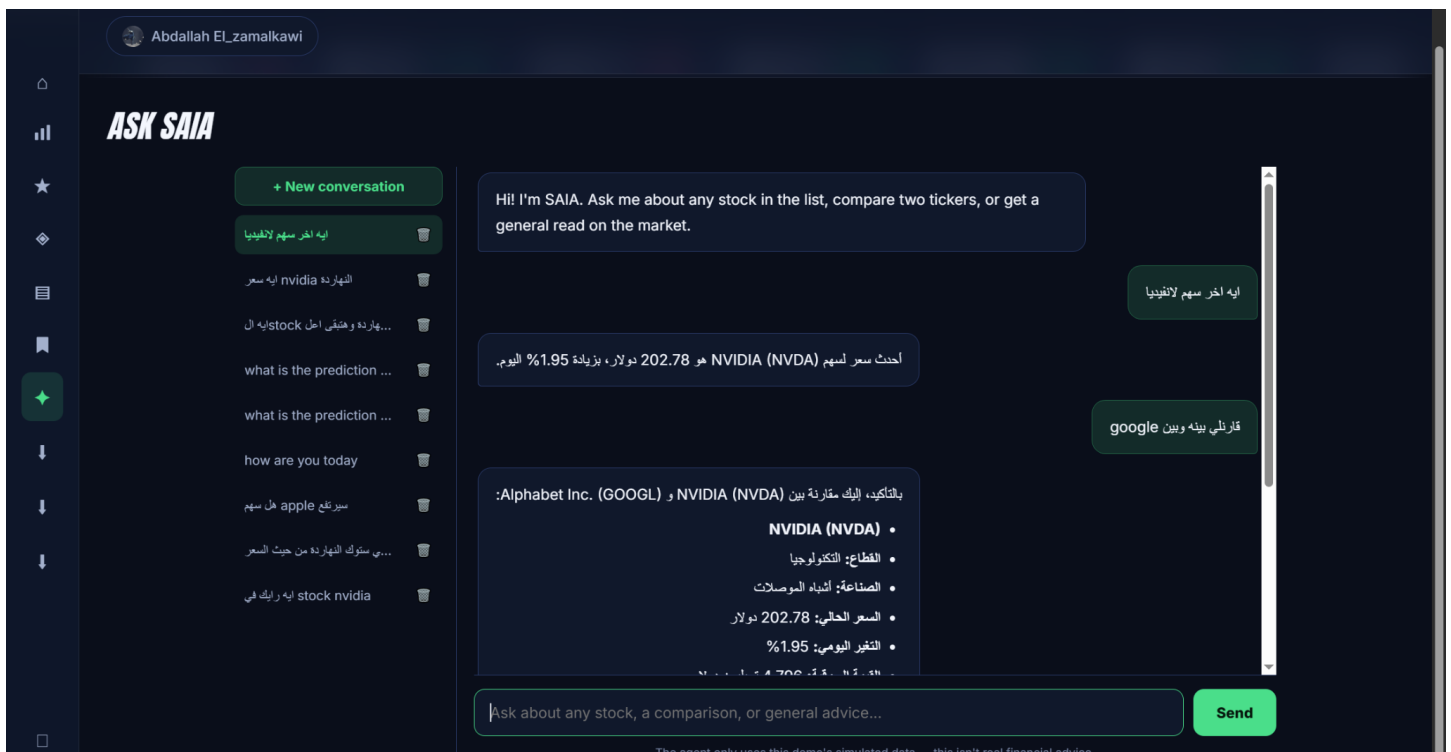
4.3.3 SAIA Conversational AI Agent (LangGraph + Gemini)

SAIA is a LangGraph agent — a small state graph that alternates between an "agent" node (Gemini, with a fixed thinking budget for consistent tool selection) and a "tools" node that executes whichever function calls Gemini requests, feeding results back until the agent has enough information to answer. Conversation state is persisted per thread_id (the user's email, or a session id for anonymous visitors) using a SQLite-backed checkpointer, so multi-turn conversations survive across requests without the frontend having to resend history; a token-budget trim keeps each Gemini call bounded regardless of how long the stored conversation gets.

The agent is bound to more than twenty tools that wrap the platform's own backend logic rather than letting the model answer from memory — covering company lookup, price history, latest and keyword-searched news, top/worst predictions,

cross-ticker comparison, sector and exchange overviews, sentiment summaries, 52-week and P/E screening, volatility, a historical what-if investment-return calculator, watchlist management, and small utilities (arithmetic, current date/time). The system prompt is explicit that any factual figure — a price, a percentage, a prediction, a sentiment score — must come from a tool call made in that same turn, never from memory or an earlier turn's answer, and that model outputs like predictions and sentiment scores must be presented as such rather than as fact or advice.

Operationally, the module supports multiple Gemini API keys with automatic fail-over on quota, permission, or authentication errors (while genuine bugs are left to surface rather than masked by a key switch), and is intentionally decoupled from the web framework — it exposes a single `ask_chatbot()` function that a REST endpoint, CLI, or websocket could all call — with the backend module imported lazily to avoid a circular import.



4.4 Training, Tuning and Validation Strategy

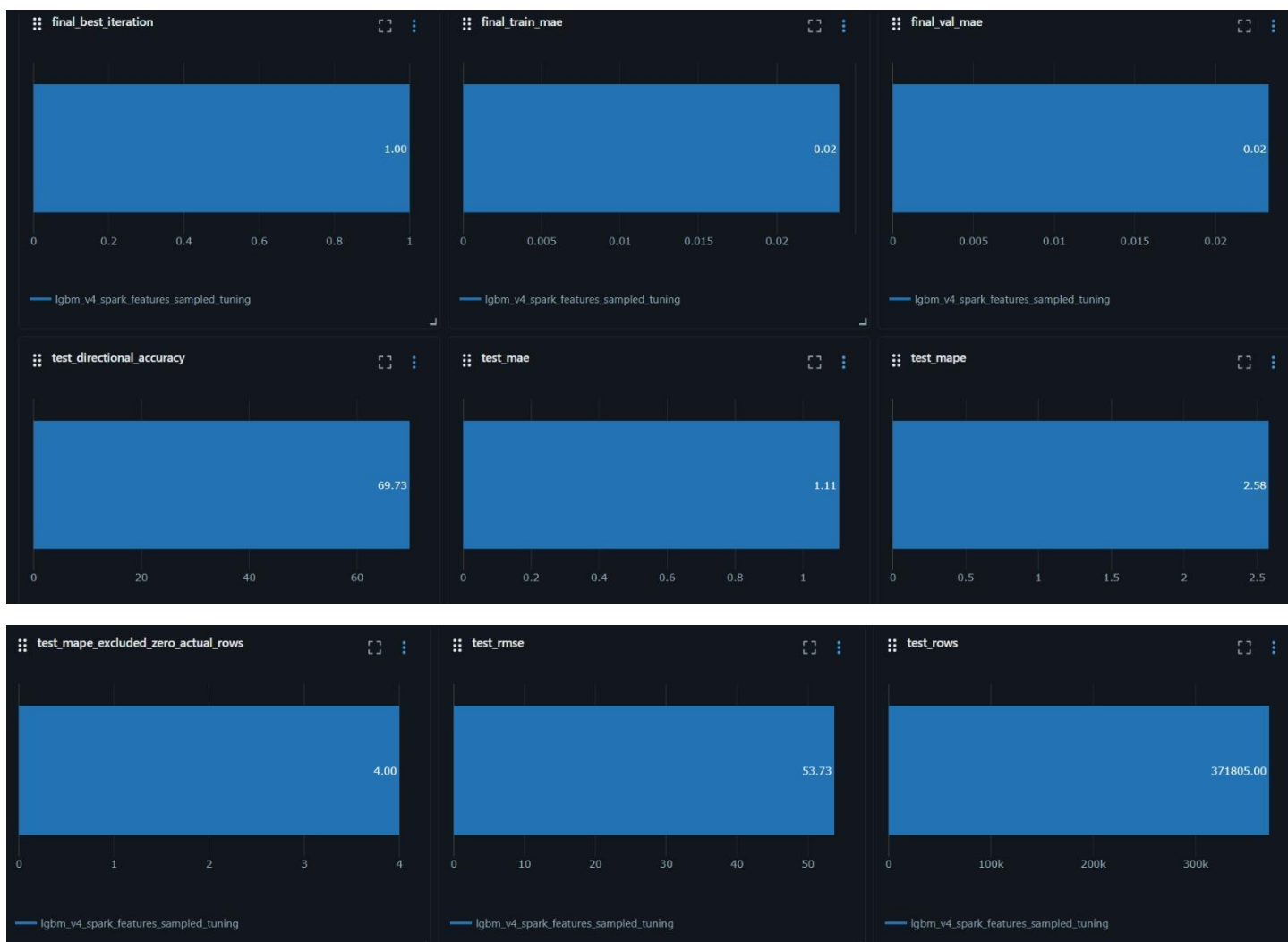
The price-prediction data is split by trading date, not by row, into three consecutive blocks — 70% train, 15% validation, 15% test — so the model is always evaluated on dates strictly after anything it was trained on, avoiding look-ahead bias. The validation split is used both for Hyperopt's search objective (minimizing validation MAE) and for early stopping when training the final model; the test split is never touched during training or tuning.

Every run is logged to MLflow: configuration parameters (lookback window, lag horizons, rolling windows, tuning-sample size), the winning hyperparameters, and the trained booster itself as a model artifact with an input-example signature. A defensive check closes out any orphaned "RUNNING" run left behind by a previous failed execution before a new run starts, since Databricks notebook cells cannot share a single try/finally across cells.

4.5 Evaluation Metrics and Business Justification

Final metrics are computed only on the held-out test set, never on the validation set the model was implicitly tuned against, to give an unbiased read of real-world performance:

- Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) in price terms, reconstructed from predicted log-returns applied to the last known adjusted close.
- Mean Absolute Percentage Error (MAPE), with rows that have a zero actual price excluded from the calculation (and reported separately) so a single bad data point cannot silently poison the average.
- Directional accuracy — whether the predicted move's sign matches the actual move's sign — with the platform's own guidance noting that a directional accuracy consistently above roughly 68–70% on a genuinely held-out set is considered a meaningful trading signal in financial forecasting, where near-random performance is the norm.



For the sentiment engine, business justification rests on FinBERT's domain fit (fine-tuned specifically on financial text, versus a general-purpose sentiment model) and on the Delta MERGE design, which makes re-scoring or backfilling safe without creating duplicate signals for downstream models or the chatbot.

4.6 Results and Business Insights

- Moving feature engineering from Pandas to native Spark window functions removed the single largest bottleneck in the prediction pipeline, with no change in feature definitions or model behavior.
- Forward as-of mapping of news to the next trading date ensures weekend and holiday coverage still reaches the model, instead of being silently dropped by a same-day join.

- Sampled hyperparameter tuning keeps search cost roughly constant even as the tracked universe grows well past ten thousand tickers, without compromising the final model, which is always retrained on full data.
- A single global, scale-free model generalizes across tickers of very different price magnitudes without needing per-ticker retraining or maintenance.
- The conversational agent turns the platform's existing dashboards and tables into a natural-language interface, letting non-technical users get grounded, tool-verified answers instead of navigating filters — while explicit grounding rules keep it from ever inventing a number.

4.7 System Architecture and Component Integration

The three components form a pipeline, not three isolated systems. Bronze news lands first; the sentiment engine cleans and scores it into `silver.news_sentiments`. The price-prediction pipeline reads that same silver sentiment table alongside `silver.stock_prices`, engineers features in Spark, trains/predicts, and writes `gold.stock_price_predictions` atomically (with scratch tables dropped in a finally block even if the write fails). The chatbot sits on top of all of it: its tools call the same backend functions that serve the platform's REST API and dashboards, so a number the agent reports is guaranteed to match what a human would see by querying the same tables directly — there is exactly one source of truth, read three different ways.

4.8 Deployment and Operational Considerations

- Incremental loads are watermark-driven off `MAX(dwh_loaded_at)` already stored in the target table itself, so no separate tracking table is needed and reruns are naturally idempotent via MERGE.
- Delta tables are periodically optimized and Z-ordered on the columns most queries filter or sort by (ticker, `published_at` / `trade_date`) to keep both the pipelines and the chatbot's tool calls fast.

- Short-lived in-process caches (about 60 seconds) sit in front of the largest lookup tables the chatbot queries, so a single multi-tool-call conversation turn doesn't re-hit the backend repeatedly.
- The chatbot's multi-key Gemini configuration and SQLite-backed conversation memory are both aimed at production resilience — automatic key fail-over on quota/auth errors, and conversation continuity that survives process restarts.

4.9 Limitations

- FinBERT sentiment scores and LightGBM price predictions are model outputs, not guarantees or financial advice; the chatbot is explicitly instructed to caveat them as such whenever it presents them.
- A single global model trades some per-ticker specialization for scalability and maintainability; an individual ticker with unusual, idiosyncratic behavior may be modeled less precisely than a dedicated single-ticker model could achieve.
- Hyperparameter search runs on a sampled subset of tickers for cost reasons; the resulting hyperparameters, while validated on the full dataset during final training, are not separately re-tuned per ticker.
- Very low-priced or thinly-traded tickers are deliberately filtered out of percentage-based screening (movers, biggest gainers/losers) to avoid noise, which means some legitimate small-cap moves are intentionally hidden from that particular view.
- The conversational agent depends on an external LLM provider (Gemini); its availability, latency, and quota are outside the platform's direct control, mitigated but not eliminated by multi-key fail-over.